LESSON PLAN

AIRCRAFT HUSBANDRY, CORROSION PREVENTION AND SURFACE PROTECTION (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various locking devices, Fasteners and fork joints
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various locking devices, Fasteners and fork joints
5. Specific Objectives	25.1.1.2 Trainee will be able to explain the locking and swaging etc.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of locking devices and fasteners; Safety precautions regarding tool handling; Instructor supervision during identification exercises
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of aircraft fasteners?
(b) Name two common types of locking devices used in aviation.
(c) Why is corrosion prevention critical for structural integrity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of locking devices • Function of split locks and safety wires • Application of lock washers and nylon inserts • Standard fastener grades and materials • Selection criteria for specific applications	L & Ds	PPT, IB	15 min
(b)	• Fork joint construction and assembly • Torque specifications for fasteners • Inspection criteria for worn fasteners • Procedures for reusing locking devices • Documentation requirements for maintenance	L & Ds	PPT, PDF	20 min
(c)	• Common failure modes in locking devices • Preventive measures for corrosion control • Troubleshooting fastener installation issues • Best practices for torque application • Record keeping for fastener replacement	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a split lock and a safety wire.
(b) Describe the procedure for inspecting a fork joint for wear.
(c) What are the key indicators that a fastener must be replaced?

12. Home Assignment and Follow Up	Prepare a brief report on the types of locking devices covered in this lesson and their specific applications in aircraft structures, referencing the next lesson on swaging, plumbing and flaring processes.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Swaging, plumbing and flaring process and tools used	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min